

TOPIC 5: CHROMOSOMAL THEORY OF INHERITANCE, LINKAGE & RECOMBINATION

PU-II Principles of Inheritance and Variation • Topic-Wise Worksheet

Student Name: _____ Roll / Reg. No: _____ Date / Batch: _____

SECTION A: Multiple Choice Questions (1 Mark Each)

- The Chromosomal Theory of Inheritance was proposed in 1902 by:
(A) Gregor Mendel (B) Walter Sutton and Theodore Boveri
(C) T.H. Morgan (D) Watson and Crick
- Who is famously known as the 'Father of Experimental Genetics' for his work on *Drosophila*?
(A) Alfred Sturtevant (B) T.H. Morgan
(C) Carl Correns (D) Hugo de Vries
- In Morgan's cross between yellow body-white eye females and wild males, the recombinant frequency was:
(A) 1.3% (B) 37.2%
(C) 50% (D) 98.7%
- One map unit (1 CentiMorgan) corresponds to what percentage of crossing over between linked genes?
(A) 0.1% (B) 1%
(C) 10% (D) 50%

SECTION B: Fill in the Blanks (1 Mark Each)

- Mendel's findings remained unrecognised until 1900 when they were rediscovered by De Vries, Correns, and
- The physical association of two or more genes on the same chromosome is termed
- The generation of non-parental gene combinations during meiosis is called
- Genetic maps based on recombination frequency were first constructed by Morgan's student

SECTION C: Very Short Answer Questions (2 Marks Each)

- [2M] List two key parallels between Mendelian genes and chromosomes during cell division.

- [2M] Define 'Linkage' and 'Recombination'.

SECTION D: Short Answer Questions (3 Marks Each)

- [3M] Why did T.H. Morgan select fruit fly (*Drosophila melanogaster*) for his experimental studies? State three reasons.

12. [3M] Explain how the physical distance between linked genes affects the strength of linkage and frequency of crossing over.

SECTION E: Long Answer / Essay Questions (5 Marks Each)

13. [5M] Describe the Chromosomal Theory of Inheritance. Explain how Sutton and Boveri united Mendelian principles with chromosomal segregation during meiosis.

14. [5M] Describe T.H. Morgan's dihybrid crosses in *Drosophila* showing tight linkage (body colour & eye colour) vs loose linkage (eye colour & wing size). Explain the deviation from 9:3:3:1 ratio.